Deep Learning for Business Made Simple

Dr. Chester Ismay

Elevator Pitch: Deep Learning for Business, Made Practical

- A hands-on, business-first introduction to deep learning -- no heavy math required
- Build and interpret neural networks for images, text, and structured business data with Keras 3 (PyTorch backend)
- Use Generative AI to write code, debug models, and explain results
- Leave able to spot where deep learning creates real business value -- and communicate it

Meet your Instructor

Dr. Chester Ismay (please call me “Chet”)

- Data science educator and AI consultant
- PhD in Statistics
- Co-author of “Statistical Inference via Data Science: A ModernDive into R and the Tidyverse” (moderndive.com/v2/)
- Background across academia, online education, corporate training, bootcamps, and consulting
- Been to all 50 US States

Learning Objectives

- By the end of this course, you will be able to:
  - Understand deep learning's building blocks and their business uses
  - Build, train, and evaluate neural networks with Keras 3 (PyTorch backend)
  - Prepare business data with preprocessing and feature extraction
  - Use Generative AI to assist with coding, debugging, and interpretation
  - Explain and visualize results for non-technical audiences
  - Recognize when deep learning offers a strategic advantage

Course schedule

- Intro: Deep Learning in the Modern Business Landscape
- Module 1: Demystifying Neural Networks for Business Insight
- Module 2: Deep Learning for Images and Customer Experience
- Module 3: Deep Learning for Text and Customer Sentiment
- Module 4: Interpreting and Communicating Deep Learning Results
- Wrap-Up and Review

Poll in ON24 (to be entered by O’Reilly team)

Which deep learning application are you most eager to explore today?

- A. Image classification and visual insights
- B. Text and customer sentiment
- C. Predicting outcomes from business data
- D. Interpreting and communicating model results

Intro:
Deep Learning in the Modern Business Landscape

What Is Deep Learning?

- Neural networks that learn patterns directly from raw data -- images, text, and numbers
- A branch of machine learning that learns its own features instead of being hand-engineered
- The engine behind image recognition, language understanding, and recommendations
- The AI chat assistants you already use -- large language models (LLMs) -- are deep learning, too

IMAGE PROMPT

Side-by-side: 'Traditional ML' (a person hand-crafting features from data) vs 'Deep Learning' (the network learns features automatically). Flat icons, blue/teal.

Why Deep Learning Now?

- More data than ever from apps, sensors, and customers
- Accessible compute -- a laptop or a free cloud notebook is enough to start
- High-level frameworks (Keras 3, PyTorch) remove the math-heavy barrier

IMAGE PROMPT

Three rising arrows labeled 'Data', 'Compute', and 'Friendly tools' converging on a glowing neural-network icon. Optimistic, blue/teal flat style.

Where Deep Learning Creates Business Value

- Customer insight -- sentiment from reviews and support logs
- Product recommendation and personalization
- Trend and demand forecasting
- Automation -- image tagging, document processing, defect detection

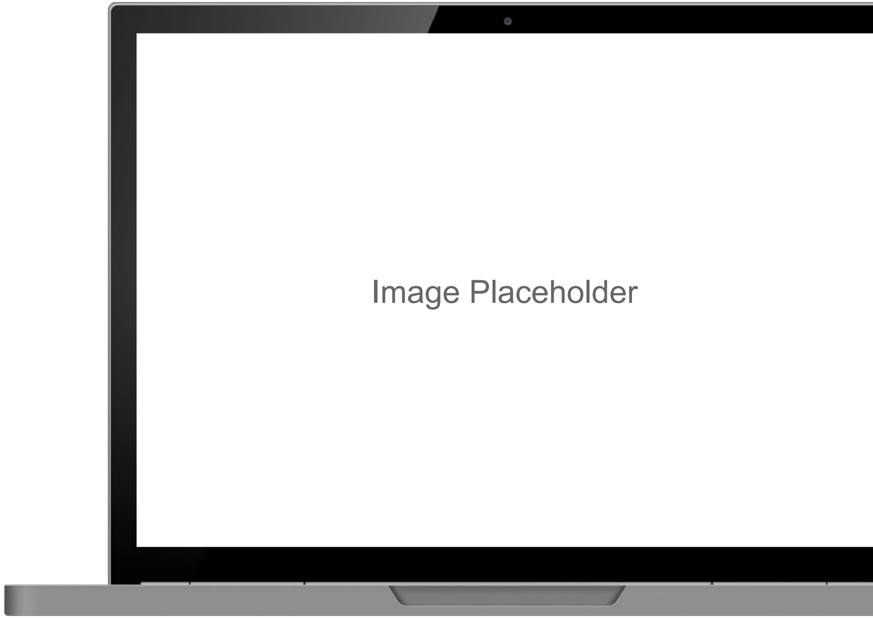
IMAGE PROMPT

2x2 grid of clean business icons: speech bubble (sentiment), cart with a star (recommendation), upward trend line (forecasting), camera with a checkmark (automation). Flat, blue/teal.

Exercise #1

Setting Up the Python Environment with Keras 3, PyTorch, and Colab

- By the end of this exercise, you will be able to:
    - Open and configure a Google Colab / Jupyter notebook for deep learning
    - Install and import Keras 3 (PyTorch backend)
    - Verify the environment with a quick tensor operation
    - Use a GenAI assistant to resolve setup or import errors
- Tip: follow along in the live notebook -- we will screen-share it.



Q&A

Intro

Module 1:
Demystifying Neural Networks for Business Insight

What Is a Neural Network?

- A neuron is a simple decision-maker that weighs its inputs
- A layer is a team of specialists, each spotting different patterns
- Stacked layers turn raw inputs into a useful prediction

IMAGE PROMPT

Friendly diagram: inputs flow into a layer drawn as a team of specialist workers, then on to an output. Flat, approachable, blue/teal.

How Neural Networks Learn

- Training is an iterative loop: predict, measure the error, adjust
- Each pass over the examples nudges the model to do a little better
- Depth lets the network learn increasingly abstract concepts
- Scale these same ideas to billions of connections trained on massive text, and you get an LLM

IMAGE PROMPT

A loop labeled 'predict -> measure error -> adjust' wrapping a small network, with a dial moving from 'wrong' toward 'right'. Blue/teal flat.

How Neural Networks "Remember"

- Knowledge is spread across the whole network, not stored at one address
- Recall works from partial or fuzzy cues -- like remembering a face from a name
- Robust to noise and small errors, so messy real-world inputs still work
- Unlike a database, there is no exact key to look up

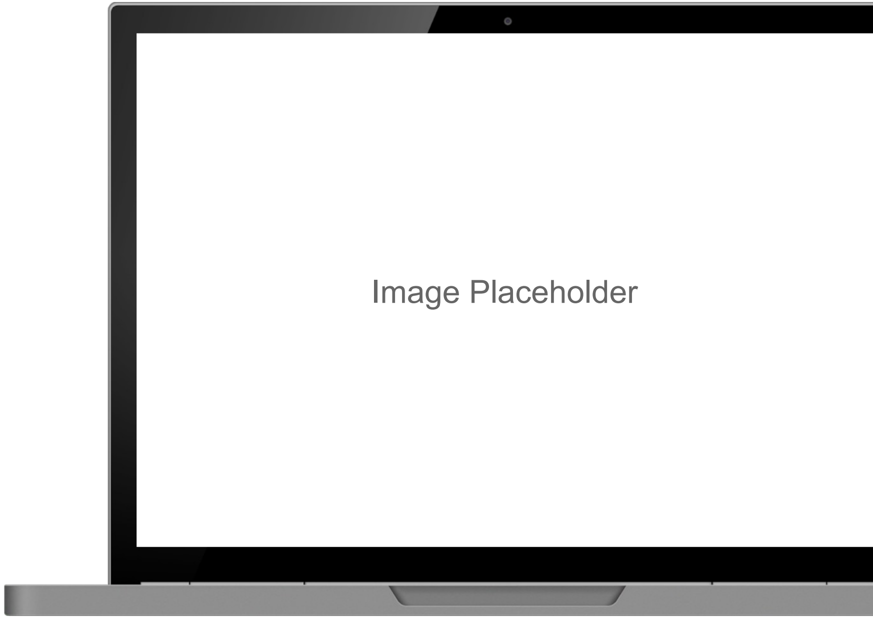
IMAGE PROMPT

Associative memory idea: a partial or blurry input (a half-seen product photo, a typo'd review) still triggers the right recall across a distributed network. Blue/teal.

Exercise #2

Building a Simple Feedforward Neural Network in Keras (PyTorch backend)

- By the end of this exercise, you will be able to:
  - Build a feedforward (dense) network with the Keras Sequential API
  - Predict a business outcome such as a booking cancellation or price
  - Compile the model with a loss function and optimizer, then fit it
  - Read the training output to gauge whether the model is learning
- Tip: follow along in the live notebook -- we will screen-share it.



Model Components, Part 1: Layers and Activations

- Layers and neurons give the model its capacity to learn patterns
- Activation functions let it capture non-linear, real-world relationships
- More layers means more capacity -- and more risk of overfitting

IMAGE PROMPT

Labeled anatomy of a small neural network: input, two hidden layers, output, with a callout 'activation = adds non-linearity'. Clean, blue/teal.

Model Components, Part 2: Loss and Optimization

- The loss function is the business-aligned definition of “wrong”
- The optimizer and learning rate control how fast and stably it improves
- The real goal: low error on NEW data, not just the training set

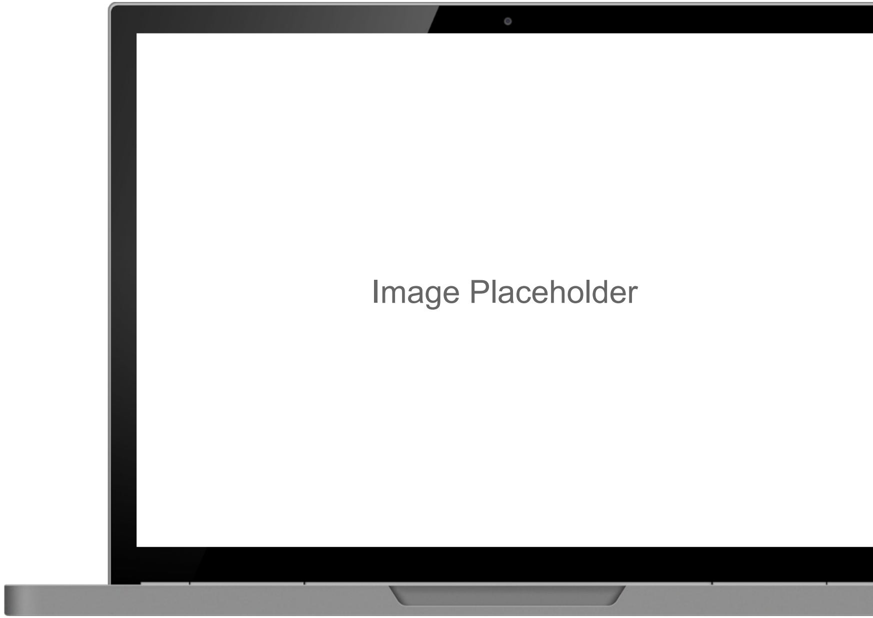
IMAGE PROMPT

A ball rolling downhill toward the lowest point of a curve labeled 'error', with a small dial for 'learning rate'. Simple, blue/teal flat.

Exercise #3

Using Generative AI to Interpret Keras and PyTorch Errors and Clarify Model Behavior

- By the end of this exercise, you will be able to:
    - Paste a Keras or PyTorch error with context and get a plain-English explanation
    - Ask a GenAI assistant to explain a layer or a hyperparameter choice
    - Iterate prompts to debug shape mismatches and dtype issues
    - Validate the suggested fix by re-running the cell
- Tip: follow along in the live notebook -- we will screen-share it.



Q&A

Module 1

Break

5 minutes

Module 2:
Deep Learning for Images and Customer Experience

How CNNs See: From Pixels to Patterns

- Convolutional neural networks (CNNs) specialize in images
- They build understanding in stages: edges, then shapes, then whole objects
- Features are learned automatically -- no hand-written rules per product

IMAGE PROMPT

A product photo on the left feeding into a CNN whose layers visualize edges -> textures -> object parts -> label. Show the feature hierarchy simply, blue/teal flat style.

CNNs in Business

- Product image classification and automatic tagging
- Defect detection in manufacturing and quality control
- Visual brand monitoring across social and retail imagery

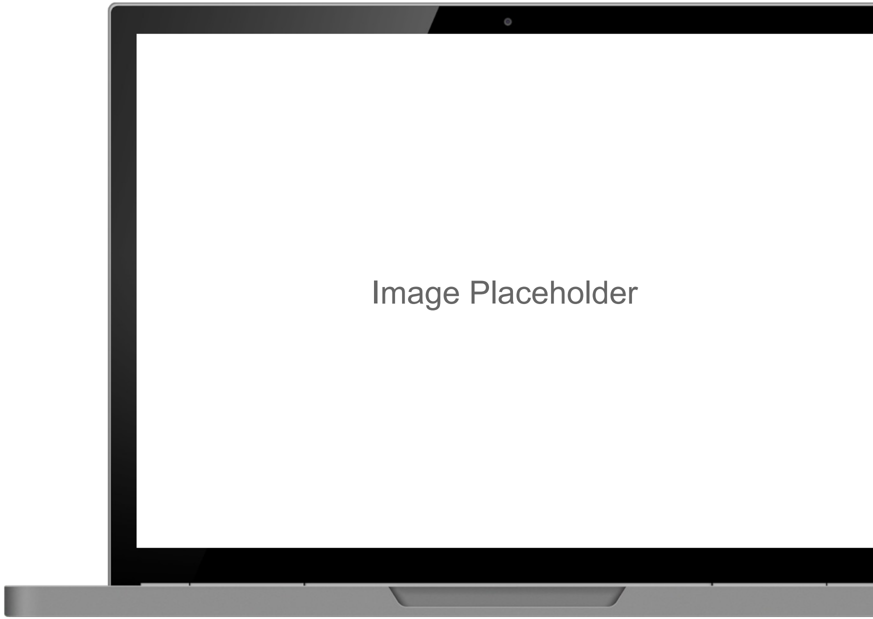
IMAGE PROMPT

Three small scenes: a tagged product photo, a flagged defective item on a line, and a brand logo spotted in social images. Clean icons, blue/teal.

Exercise #4

Classifying Product Images Using a Pre-Trained CNN in Keras

- By the end of this exercise, you will be able to:
    - Load a pre-trained CNN (for example MobileNet or ResNet) from Keras
    - Preprocess and classify sample product images
    - Interpret the top predicted labels and their confidence
    - Discuss business uses of automated image tagging
- Tip: follow along in the live notebook -- we will screen-share it.



Transfer Learning: Reuse, Don’t Reinvent

- Start from a model already trained on millions of images
- Keep its general visual knowledge; retrain only a small new part
- Fine-tune on your own, much smaller, business dataset

IMAGE PROMPT

A large pre-trained model with most layers locked and a small new head being trained on a tiny company dataset; arrows show knowledge transfer. Blue/teal flat.

Why Transfer Learning Matters for Business

- Far less data and compute needed -- faster time to value
- Makes deep learning feasible for smaller organizations
- Lowers the cost and risk of trying an AI project
- LLMs are the ultimate example -- a foundation model pre-trained once, then adapted to many tasks

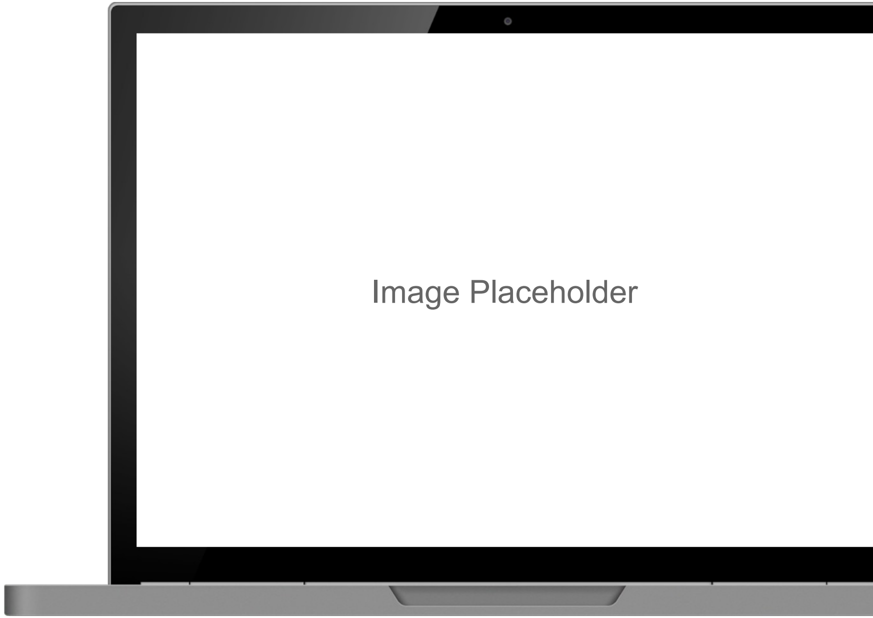
IMAGE PROMPT

A small team reaching a high shelf by standing on a giant 'pre-trained model' block -- the 'stand on the shoulders of giants' idea. Warm, blue/teal.

Exercise #5

Using Generative AI to Explain Model Layers and Suggest Architecture Improvements

- By the end of this exercise, you will be able to:
  - Ask a GenAI assistant to summarize what each layer of a model does
  - Request architecture suggestions for your specific use case
  - Compare trade-offs: accuracy vs. model size vs. speed
  - Apply and test one suggested change
- Tip: follow along in the live notebook -- we will screen-share it.



Q&A

Module 2

Break

5 minutes

Module 3:
Deep Learning for Text and Customer Sentiment

Teaching Machines to Read

- Turn unstructured text -- reviews, tickets, chats -- into business signal
- Move from counting keywords to understanding meaning and intent
- Handle the messy reality of how customers actually write

IMAGE PROMPT

A pile of messy customer text (reviews, emails, chat bubbles) funneling into a tidy 'signal' meter. Clean, blue/teal flat.

Word Embeddings: Meaning as Numbers

- Words become points in space, so a model can do math on meaning
- Words with similar meaning sit close together
- Captures context -- “not great” reads very differently from “great”

IMAGE PROMPT

A 2D map where related words cluster: 'refund / return / cancel' in one group, 'love / great / excellent' in another. Blue/teal.

Large Language Models (LLMs): NLP at Scale

- LLMs are deep learning for language, trained on enormous amounts of text
- They build on embeddings to predict the next word, again and again, into fluent responses
- This is what powers chat assistants, summarization, translation, and Q&A
- For business: draft content, summarize feedback, and answer questions over your own documents

IMAGE PROMPT

A simple flow: a mountain of text -> a large neural network (the LLM) -> it predicts the next word to produce fluent answers, summaries, and drafts. Friendly, blue/teal.

NLP in Business

- Customer sentiment from reviews and surveys
- Triage and themes from support logs and tickets
- Brand and product feedback monitored at scale

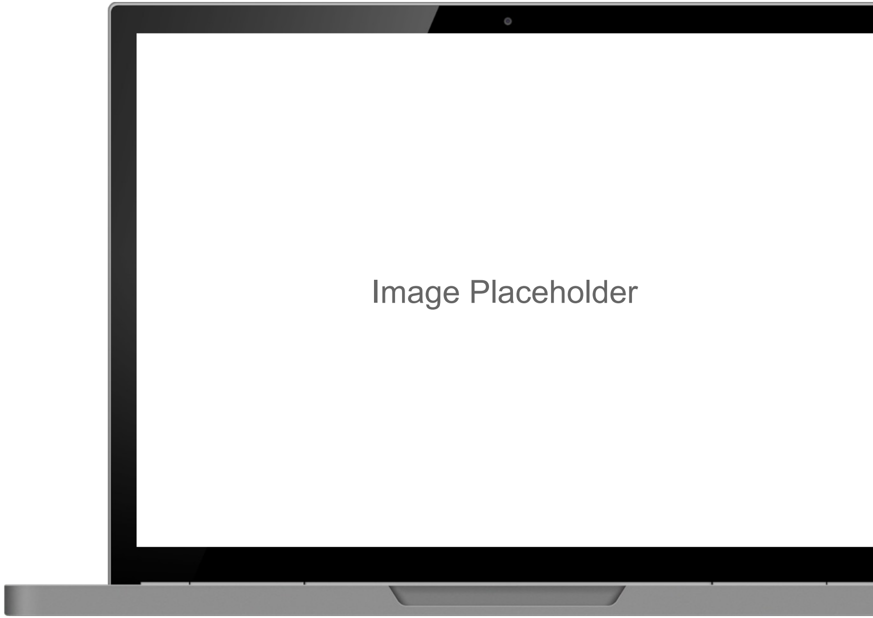
IMAGE PROMPT

A dashboard showing a sentiment trend line, a topic word-cloud, and a support-ticket count. Executive-friendly, blue/teal.

Exercise #6

Building a Sentiment Analysis Model with Embeddings in Keras (PyTorch backend)

- By the end of this exercise, you will be able to:
  - Prepare and tokenize text data
  - Add an embedding layer to a Keras model
  - Train the model to classify sentiment (positive vs. negative)
  - Evaluate performance on held-out reviews
- Tip: follow along in the live notebook -- we will screen-share it.



Traditional Analytics vs. Deep Learning for Text

- Rule and keyword methods: fast and transparent, but brittle
- Deep learning: captures context, nuance, and even sarcasm
- You can also prompt a pre-trained LLM -- often no training data or model-building needed
- Choosing the right tool: cost, data volume, and interpretability

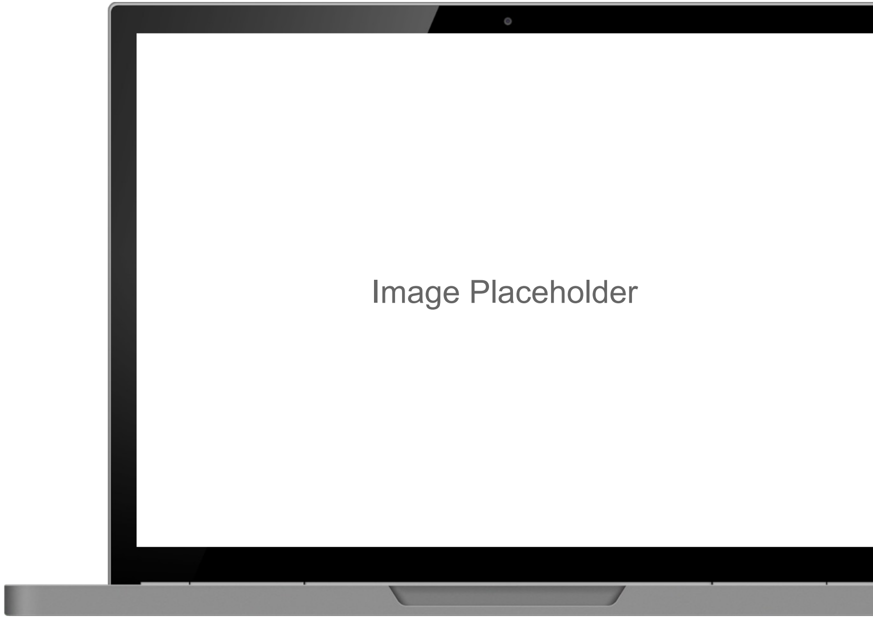
IMAGE PROMPT

Balanced comparison: left panel 'Keyword/rules' (magnifying glass over a word list), right panel 'Deep learning' (a network understanding a full sentence). Neutral, blue/teal flat.

Exercise #7

Using Generative AI to Summarize Sentiment Results into Stakeholder-Ready Narratives

- By the end of this exercise, you will be able to:
    - Feed model outputs and metrics to a GenAI assistant
    - Generate a concise, plain-English summary
    - Tailor the tone for executives versus analysts
    - Fact-check and refine the narrative
- Tip: follow along in the live notebook -- we will screen-share it.



Q&A

Module 3

Break

5 minutes

Module 4:
Interpreting and Communicating Deep Learning Results

Beyond Accuracy: Metrics That Matter

- Judge models on unseen data -- one that memorizes the training set fails in production
- Accuracy alone misleads, especially with imbalanced data
- Use precision, recall, and F1 to see what the model gets right and wrong

IMAGE PROMPT

A confusion matrix with precision and recall callouts, beside a small 'train vs. new data' split showing overfitting. Clean, blue/teal.

Aligning Metrics with Business Costs

- Match the metric to the cost of each type of error
- Tune the decision threshold to the business trade-off
- Watch for class imbalance and data leakage

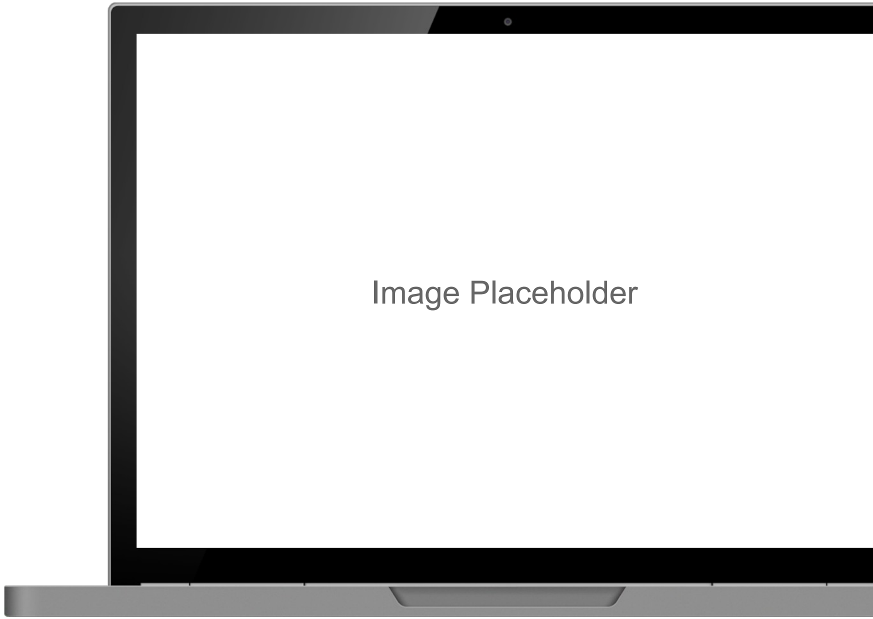
IMAGE PROMPT

A balance scale weighing the cost of a false positive vs. a false negative, with small dollar-sign tags. Clear, blue/teal.

Exercise #8

Interpreting Predictions and Visualizing Performance with Confusion Matrices and Charts

- By the end of this exercise, you will be able to:
  - Generate predictions on validation data
  - Build and read a confusion matrix
  - Visualize performance with clear, simple charts
  - Translate metrics into business implications
- Tip: follow along in the live notebook -- we will screen-share it.



Opening the Black Box

- Show “how the model thinks” with concrete examples
- Feature importance and attention maps highlight what drove a prediction
- Turn model internals into reasons a stakeholder can follow

IMAGE PROMPT

An explainability view: one prediction highlighted with a short 'why' list of top contributing factors. Uncluttered, blue/teal.

Telling the Story for Non-Technical Audiences

- Lead with the decision and its impact, then support it with evidence
- Prefer visuals over jargon and raw numbers
- Use AI-assisted summaries to draft a clear executive brief

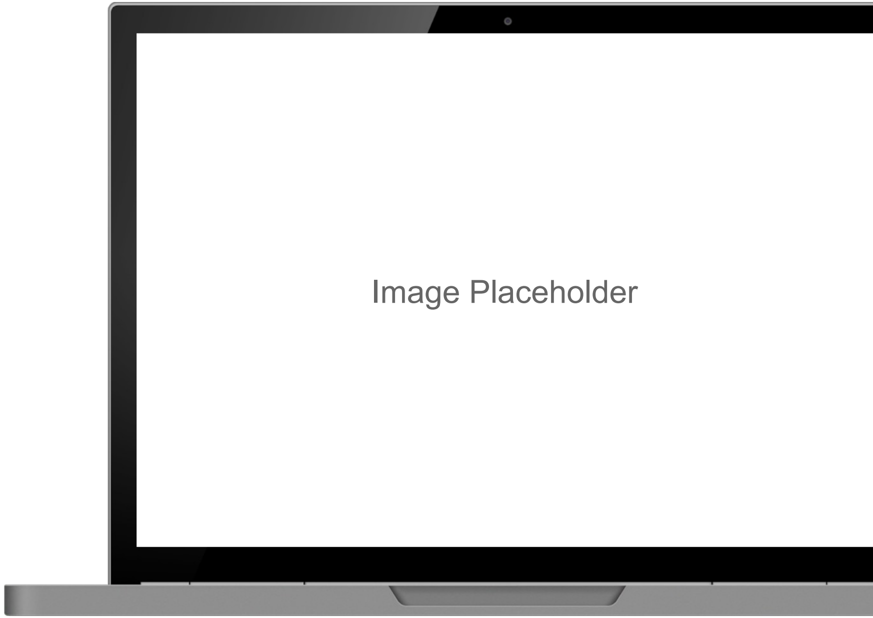
IMAGE PROMPT

A presenter showing a simple, confident chart to executives: one headline number and a short takeaway. Warm, blue/teal.

Exercise #9

Using Generative AI to Generate Executive Summaries of Model Outcomes

- By the end of this exercise, you will be able to:
    - Summarize model results into a short executive brief
    - Highlight business impact, risks, and recommended actions
    - Generate a slide-ready narrative
    - Review the output for accuracy and bias
- Tip: follow along in the live notebook -- we will screen-share it.



Q&A

Module 4

Wrap-Up and Review
Bringing It All Together

Deep Learning Myths vs. Reality (1 of 2)

- “It needs heavy math/CS” -- frameworks abstract most of the complexity
- “Neural nets are black boxes” -- explainability tools make them interpretable
- “Bigger models are always better” -- beware overfitting and diminishing returns

IMAGE PROMPT

Left column of muted myth speech-bubbles vs a right column of green check-mark realities (first three myths). Balanced, blue/teal accents.

Deep Learning Myths vs. Reality (2 of 2)

- “Deep learning finds insights on its own” -- human and domain expertise are essential
- “Good prompting is just typing a question” -- it is a skill: clarity, specificity, iteration
- “LLMs know what is truth” -- they predict plausible text, so verify facts and sources
- “You need GPUs and big data” -- transfer learning makes it accessible

IMAGE PROMPT
Continuation: more muted myth speech-bubbles vs green check-mark realities. Balanced, blue/teal accents.

Discussion: From Myths to Your Workflow

- Which deep learning myth surprised you most -- and why?
- Where in your organization could images, text, or structured-data models add value?
- Share one idea in the chat; we'll discuss a few together.

Takeaways

- Build, train, and read deep learning models with Keras 3 (PyTorch backend)
- Transfer learning makes deep learning practical on modest data and hardware
- Generative AI assists coding, debugging, and explanation -- always validate it
- The value is in interpretation and communication, not just accuracy
- Keep going:
  - Keras 3, PyTorch, and Google Colab documentation
  - Book: Deep Learning with Python, Third Edition (Chollet)
  - Book: Deep Learning for Coders with fastai and PyTorch
  - For the mathematically curious (free): D. MacKay, Information Theory, Inference, and Learning Algorithms -- inference.org.uk/itprnn/book.pdf

Thank you!

O'REILLY®

Integrate deep learning into your organization's analytics practice.

Dr. Chester Ismay · chester.ismay@gmail.com · https://chester.rbind.io